Hiroki SAWADA

Phone: (+81)-80-6787-0693
Private Email: hiroki.sawada.research@gmail.com
Official Email: hiroki.sawada@csl.sony.co.jp
Personal Page: https://www.hiroki-sawada.com

EDUCATION

2016.4 - 2020.3 **Keio University**
BA in Applied Physics and Physico-Informatics
Kanagawa, Japan

2020.9 – 2026.6 **Okinawa Institute of Science and Technology Graduate University**
Ph.D. in Computer Science – Cognitive Neuro-robotics Research Unit
Okinawa, Japan

RESEARCH EXPERIENCE

2026.4 - present **Sony Computer Science Laboratory, Humanity Science Project**
Project Researcher
Tokyo, Japan
Body Integration Learning Framework for Highly Skilled Robots

- Leading the setup of physical human-robot interaction experiments using the open-source humanoid robot “OpenArm” (Enactic).
- Proposed PredVLA, a compact recurrent controller organized around predicting sensorimotor dynamics and inferring its latent state from prediction error, rather than compressing a conventional observation-to-action policy[1].

2025.1 – 2025.7 **Équipes Traitement de l’Information et Systèmes Lab**
CY Cergy Paris University
Prof. Alexandre Pitti, Prof. Mathias Quoy
Cergy, France
Investigation of Robust Predictive-Coding inspired Recurrent Neural Network in Real Robot Tasks

- Proposed CERNet, a hierarchical predictive-coding RNN with class embedding, enabling unified real-time motion generation, recognition, and meta-inference of confidence.
- Demonstrated stable reproduction of 26 handwritten alphabet trajectories and confidence estimation using the Reachy humanoid robot (Pollen Robotics) in both simulation and real-world settings.
- Accepted for *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA), 2026* [3].

2021.9 – 2026.6 **Cognitive Neuro-robotics Research Unit**
Okinawa Institute of Science and Technology Graduate University
Prof. Jun Tani
Okinawa, Japan
Human-Robot Kinaesthetic Interaction Based on the Free Energy Principle

- Investigated human-robot kinaesthetic interaction using a predictive coding-inspired variational RNN, guided by the Free Energy Principle, to model bidirectional sensorimotor dynamics between humans and a humanoid robot.
- Through mutually adaptive physical behaviors through active inference, conducted research on revealing the relationship between kinaesthetic perception and the internal state of the employed variational RNN [1].
- Studied the emergence of novel behaviours through incremental learning with variational RNNs [2].
- Proposed a framework where a network can continuously learn new movements through human-robot interaction by gating the learning rate based on the free energy [2].

2019.7 - 2020.1
2021.1 - 2021.4 **Quantum Materials Science Unit**
Okinawa Institute of Science and Technology Graduate University
Prof. Yoshinori Okada

Okinawa, Japan

Evaluation of magnetic properties in van del Waals compounds

- Evaluated the magnetic properties of van del Waals compounds CeTe3 and CeTe2Se (collaborated with Dr. Daichi and Yuita). Results published as a full journal paper [4].

2019.4 – 2020.3 **Matoba & Kamihara Lab**
Keio University
Prof. Masanori Matoba, Prof. Yoichi Kamihara
Kanagawa, Japan

Synthesis of layered oxysulfide to optimize thermoelectric properties

- Synthesised a polycrystalline $LaCuS_{0.5}Se_{0.5}O$ which is expected to have high thermoelectric properties for the first time.
- Evaluated the effect of cupper deficiency in the polycrystalline to its thermoelectric properties. Results published in a full journal paper [5, 6].

Working Experience

2026.3 – 2026.4 **Mercor**

Bilingual Japanese STEM Expert Writer & Editor
Tokyo, Japan

Designing scientific prompts and evaluation frameworks for AI systems

- Designing Prompt-Rubrics for evaluating LLMs in the fields of science and engineering.
- Engaged in work as both a prompt creator (Writer) and an evaluator (Editor).

2023.9 – 2026.3 **R&D Engineer**
Reazon Holdings, Inc.
Tokyo, Japan

Development of Automatic Minutes Sender Bot Using LLM and NLP

- Full time software engineer (4 days/week) with permanent employment status.
- Lead engineer for an automated meeting-minutes distribution system that integrates LLMs and NLP; authored over 95% of the code. The system is now running in production within the company.
- Built a scalable backend leveraging AWS EC2, S3, SageMaker Asynchronous Inference, Lambda, and other services to enable asynchronous inference.
- Designed and developed a RAG-based GPT agent capable of answering complex queries regarding internal company policies and regulations. Improved internal knowledge accessibility and employee self-service efficiency.

Publications and Presentations（arXiv preprint: under review）

[1] **Hiroki Sawada,** and Shunichi Kasahara. “PredVLA: Predictive Sensorimotor Modeling for Sub-Million-Parameter Robot Manipulation.” ***arXiv preprint arXiv:2608.26673*** (2026). (Under Review)

[2] **Hiroki Sawada,** and Jun Tani. "Free-Energy-Gated Plasticity for Real-Time Online Motor Learning in Physical Human-Robot Interaction." ***arXiv preprint arXiv:2608.23000*** (2026). (Under Review)

Publications and Presentations（peer-reviewed）

[3] **Hiroki Sawada,** Wataru Ohata, and Jun Tani. "Human-Robot Kinaesthetic Interaction Based on The Free Energy Principle." ***IEEE Transactions on Systems, Man and Cybernetics: Systems*** (2024).

[4] **Hiroki Sawada,** Jun Tani. “Incremental Learning in Physical Human-Robot Interaction Using Predicting Coding-inspired Variational RNNs.” ***IROS (IEEE/RSJ International Conference on Intelligent Robots and Systems****, late-breaking poster presentation,* 2024.

[5] **Hiroki Sawada,** Alexandre Pitti, Mathias Quoy. “CERNet: Class-Embedding Predictive-Coding RNN for Unified Robot Motion, Recognition, and Confidence Estimation.” *Proceedings of the* ***ICRA (IEEE/RSJ International Conference on Robotics and Automation)***, 2026.

[6] Ueta, Daichi, Riki Kobayashi, **Sawada Hiroki**, et al. "Anomalous Magnetic Moment Direction under Magnetic Anisotropy Originated from Crystalline Electric Field in van der Waals Compounds CeTe3 and CeTe2Se." J*ournal of the Physical Society of Japan* 91.9 (2022): 094706.

[7] Azuma Nobuhiko, **Sawada Hiroki**, Itoh Houji et al. "Thermal transport properties of a mixed anion layered compound, polycrystalline LaCu1-δS0.5Se0.5O (δ = 0.01)." *Korean Journal of Materials Research* 34, 464-474 (2024).

[8] Azuma Nobuhiko, **Sawada Hiroki,** Itoh Houji, et al. "In-gap-states of a mixed anion layered compound, polycrystalline LaCu1-deltaS0.5Se0.5O(delta~0.01) as a degenerate semiconductor." *Materials Science and Technology Japan* 58, 64-68(2021).

Personal Projects

2025.3 – present

LLM-based Mobile Game App (in development)
Personal Project | Swift, Python, OpenAI API

- Designing and developing an iOS game app where users ask questions to guess an answer the AI has thought of.
- Integrated OpenAI GPT API and AWS Lambda for real-time inference and data handling.
- Applied prompt engineering techniques to constrain and guide model responses across different genres (e.g., animals, brands, countries).
- Leveraged Retrieval-Augmented Generation (RAG) to reduce API call cost while maintaining consistency and relevance.
- Built entirely using Swift and Xcode as a native iOS app. Demo available upon request.

AWARDS AND HONOURS

2020.9 – 2026.6

Fully Funded Ph.D.
Okinawa Institute of Science and Technology Graduate University, Okinawa, Japan

- Selected for a fully funded Ph.D. program.

2021.4 - 2022.3

JEES Softbank AI Researcher fellowship
Softbank, Japan

- Scholarship for 1,000,000 yen without usage limitations.
- Granted to 100 Ph.D. students in the field of AI.

2025.1 - 2025. 7

Bourses France Excellence 2025
French Government, France

- Scholarship from French government to Japanese PhD students.
- Covers full tuition and monthly stipend of €1709 for 6 months.

2025.9

TOEIC Listening & Reading test: score 975 / 990

Language

English — Near Native-level fluency
(Lived 12 years abroad in an international environment; PhD program in English)

Japanese — Native

French — Conversational proficiency (CEFR A2 level)

Activities

2022.2 – 2023.2

Diversity Officer, OIST Student Council

- Promoted interaction among students from diverse cultural backgrounds within OIST.
- Organized a campus symposium “What is discrimination?”.

2023.2 – 2024.2 **Culture and External Relations Representative, OIST Student Council**

- Strengthened relationships between OIST and the local community.
- Co-hosted two sake introduction events (≈50 participants each) in collaboration with Miss Sake, an organization promoting Japanese sake worldwide.
- Organized a joint cultural event with Okinawa Prefectural University of Arts.

2025.8 **Reviewer, IEEE ICDL 2025**

- Reviewed peer-reviewed contributed papers in the field of cognitive robotics and learning for the IEEE International Conference on Development and Learning (ICDL) 2025.